Secrets of the sky

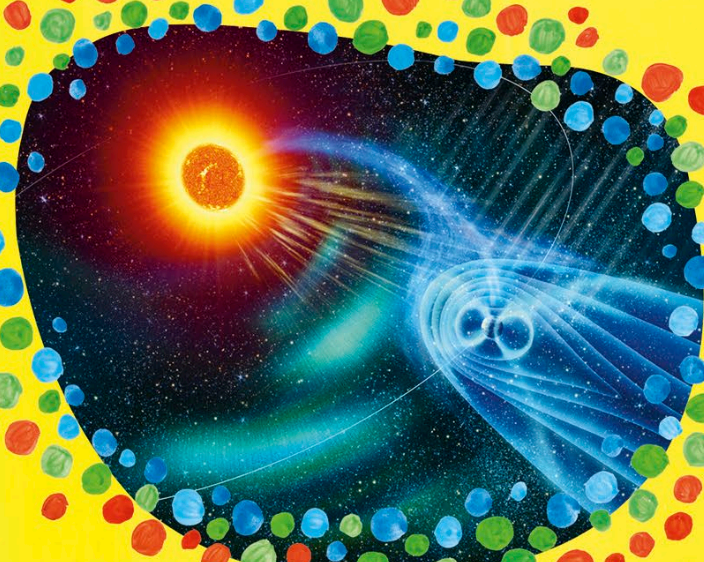
The questions Francisca likes to solve focus on the strange happenings high in the Earth's atmosphere, and deep below its surface in the mantle and core. These layers of our planet are not visible to our eyes, but physicists can use instruments to collect data, then try to spot patterns to understand what's going on.

Francisca is particularly interested in an area of the atmosphere called the ionosphere. Her work has revealed more about how Earth's climate varies over time, which has helped us realize how the things we do can affect it.

The equatorial electrojet is a strong river of electrical current high above the Earth in the ionosphere. It's caused by solar activity in the ionosphere.

Ionosphere

The ionosphere is a thick layer of Earth's atmosphere, 30 to 600 miles (50 to 1,000 km) above the ground. It's important because the radio signals used by communication and navigation systems travel through it.



The sun unleashes the solar wind—gusts of electrically charged particles that zoom through space. Francisca has studied how this "space weather" affects the equatorial electrojet.